Unit 2: First Order Linear Differential Equations — Practice Problems and Solutions

Prepared by Staples Education

Review Guide of Graded Problems with Complete Step-by-Step Solutions

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Part II — Review Guide: Practice Problems

Work the following ten problems in order; they are graded from direct separation through linear factors, initial value problems, the homogeneous substitution, and a full physical model. Solutions follow in Part III.

1. Solve $\dfrac{dy}{dx} = \dfrac{x}{y}$ by separation of variables.
2. Solve $\dfrac{dy}{dx} = e^{x-y}$.
3. Solve $\dfrac{dy}{dx} = y\cos x$, and identify any solution that is lost when you divide by y.
4. Solve the linear equation $\dfrac{dy}{dx} + 3y = 6$ using an integrating factor.
5. Solve $x\dfrac{dy}{dx} + y = x^2$ by first writing it in standard linear form.
6. Solve $\dfrac{dy}{dx} + \dfrac{2}{x}\,y = x$.
7. Solve the initial value problem $\dfrac{dy}{dx} + 2y = 4$ with $y(0) = 1$.
8. Solve the homogeneous equation $\dfrac{dy}{dx} = \dfrac{x+y}{x}$ using the substitution $y = vx$.
9. Solve the homogeneous equation $\dfrac{dy}{dx} = \dfrac{x^2+y^2}{xy}$ using $y = vx$.
10. **(Application)** An object at 100° is placed in a room held at 20°. After 10 minutes its temperature is 60°. Find the temperature function $T(t)$ and determine how long it takes the object to reach 30°.

Part III — Complete Solutions

Solution 1. Separate, then integrate.

$$y\,dy = x\,dx$$

$$\frac{y^2}{2} = \frac{x^2}{2} + C_1$$

$$y^2 = x^2 + C \qquad \text{(general solution)}.$$

Solution 2. Split the exponential as $e^{x-y} = e^x e^{-y}$, then separate.

$$e^{\,y}\,dy = e^{\,x}\,dx$$

$$e^{\,y} = e^{\,x} + C$$

$$y = \ln(e^{\,x} + C).$$

Solution 3. Separate (assuming $y \neq 0$), integrate, and exponentiate.

$$\frac{1}{y}\,dy = \cos x\,dx$$

$$\ln|y| = \sin x + C$$

$$y = C\,e^{\sin x}.$$

Dividing by y assumed $y \neq 0$, so it dropped the equilibrium solution $y = 0$ — which is recovered as the special case $C = 0$.

Solution 4. Here $P = 3$, so the integrating factor is $\mu = e^{\int 3\,dx} = e^{3x}$. Multiply through:

$$\frac{d}{dx}\left[e^{3x}y\right] = 6\,e^{3x}$$

$$e^{3x}y = \int 6\,e^{3x}\,dx = 2\,e^{3x} + C$$

$$y = 2 + C\,e^{-3x}.$$

Solution 5. Divide by x to reach standard form:

$$\frac{dy}{dx} + \frac{1}{x}\,y = x, \qquad \mu = e^{\int (1/x)\,dx} = e^{\ln|x|} = x.$$

Multiply through and integrate:

$$\frac{d}{dx}\left[xy\right] = x \cdot x = x^2$$

$$xy = \frac{x^3}{3} + C$$

$$y = \frac{x^2}{3} + \frac{C}{x}.$$

Solution 6. With $P = \frac{2}{x}$, integrate the coefficient and exponentiate:

$$\mu = e^{\int (2/x)\,dx} = e^{2\ln|x|} = x^2.$$

Multiply through and integrate:

$$\frac{d}{dx}\left[x^2 y\right] = x^2 \cdot x = x^3$$

$$x^2 y = \frac{x^4}{4} + C$$

$$y = \frac{x^2}{4} + \frac{C}{x^2}.$$

Solution 7. The integrating factor is $\mu = e^{\int 2\,dx} = e^{2x}$. Multiply through and integrate:

$$\frac{d}{dx}\left[e^{2x} y\right] = 4\,e^{2x}$$

$$e^{2x} y = 2\,e^{2x} + C$$

$$y = 2 + C\,e^{-2x}.$$

Apply the initial condition $y(0) = 1$:

$$1 = 2 + C \implies C = -1$$

$$y = 2 - e^{-2x}.$$

Solution 8. Rewrite as $\frac{dy}{dx} = 1 + \frac{y}{x}$ and substitute $y = vx$, so $\frac{dy}{dx} = v + x\frac{dv}{dx}$:

$$v + x\frac{dv}{dx} = 1 + v$$

$$x\frac{dv}{dx} = 1 \implies dv = \frac{1}{x}\,dx$$

$$v = \ln|x| + C.$$

Back-substitute $v = \frac{y}{x}$:

$$y = x\ln|x| + Cx.$$

Solution 9. Substitute $y = vx$. The right side becomes

$$\frac{x^2 + v^2x^2}{x \cdot vx} = \frac{1 + v^2}{v}.$$

With $\frac{dy}{dx} = v + x\frac{dv}{dx}$,

$$v + x\frac{dv}{dx} = \frac{1+v^2}{v} = \frac{1}{v} + v$$

$$x\frac{dv}{dx} = \frac{1}{v} \implies v\,dv = \frac{1}{x}\,dx$$

$$\frac{v^2}{2} = \ln|x| + C_1 \implies v^2 = 2\ln|x| + C.$$

Back-substitute $v = \frac{y}{x}$:

$$\frac{y^2}{x^2} = 2\ln|x| + C \implies y^2 = x^2\big(2\ln|x| + C\big).$$

Solution 10. Newton's law of cooling with surrounding temperature $T_s = 20$ and initial temperature $T_0 = 100$ gives

$$T(t) = 20 + (100 - 20)\,e^{-kt} = 20 + 80\,e^{-kt}.$$

Use the reading $T(10) = 60$ to find k:

$$60 = 20 + 80\,e^{-10k} \implies 80\,e^{-10k} = 40$$

$$e^{-10k} = \tfrac{1}{2} \implies -10k = -\ln 2 \implies k = \frac{\ln 2}{10}.$$

Hence the temperature function is

$$T(t) = 20 + 80\,e^{-(\ln 2/10)\,t} = 20 + 80\left(\tfrac{1}{2}\right)^{t/10}.$$

To reach 30°, solve $T(t) = 30$:

$$30 = 20 + 80\,e^{-kt} \implies 80\,e^{-kt} = 10 \implies e^{-kt} = \tfrac{1}{8}$$

$$-kt = -\ln 8 = -3\ln 2 \implies t = \frac{3\ln 2}{k} = \frac{3\ln 2}{\ln 2/10} = 30 \text{ minutes}.$$

Unit 2 Key Takeaway

Every method in this unit reduces to a single move — rewrite the equation into a form you can integrate. Separation of variables splits a factored rate law; the integrating factor $\mu = e^{\int P\,dx}$ collapses a linear left side into one derivative; and the substitution $y = vx$ turns a homogeneous equation separable. Read the structure first, and the technique always follows.